Business Analytics 402

Intro to SQL and Basic Data Management

Fall 2026

Prof. Ryan Rosario

Instructor Contact Information

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Office Hours:	Monday, 6:30–7:30pm in C Building Lobby Wednesday, 7:00–8:00pm on Zoom Day/time/modality/location is subject to change.

Instructor Background

Ph.D., *Statistics*, UCLA.
M.S., *Computer Science*, UCLA.
B.S. *Mathematics of Computation, Statistics*, UCLA.

I am from industry. I am originally from Thousand Oaks, CA – born and raised in Southern California and now live just outside Mammoth Lakes, CA in the Eastern Sierra.

Teaching Assistant Contact Information

Name:	Fauzan Budi Prasetya
Email:	fauzan.prasetya@ucla.edu
Office Hours:	Thursday, 11:00am–12:00pm on Zoom
Discussion:	Wednesday 12:00–1:00pm, Collins A301

About MSBA 402

This course teaches you to manage and analyze data in relational database management systems (RDBMS) using SQL: data definition, data manipulation, data control, and analytics. It covers roughly the first four weeks of Computer Science 143 (Databases), another class I teach, without the engineering topics and with additional material specific to business analytics.

This course uses PostgreSQL. Previous offerings of MSBA 402 used MySQL, but PostgreSQL is a better fit for analytics: it supports more sophisticated queries and follows the ANSI SQL standard more closely. Most of what you learn here is standard SQL that works across database systems, and I will point out PostgreSQL-specific syntax when we use it.

The structure of this course has changed. In the past, course content and assessment were based on executing day-to-day in a corporate environment; however, to get to that corporate environment, students need to pass interviews. Instead, the course is structured around content required to succeed in an interview and at work. Because of this new focus, all assessments are closed-note, closed-book, and closed-computer taken with pencil and paper.

Meeting Times

Activity	Day	Time	Location
LEC 1/2	Monday	4:10–6:00pm	Anderson Hall G313
DIS 1/2	Wednesday	12:00–1:00pm	Collins A301

Your TA will host a scheduled discussion section that reviews and practices the material covered in lecture.

Prerequisites

No previous knowledge of databases or SQL is expected. You will need a laptop for this class, and you will find it useful to bring to each class. To use PostgreSQL from your laptop, we will provide you with technology that allows you to connect to a PostgreSQL server.

Students enter this course with a wide range of experience levels. Since this class does not require a certain level of experience, we encourage a specific style for writing SQL queries, which is covered in lecture. For those learning SQL for the first time, this ensures their queries are easy for others to read and understand (an important skill in data careers).

Textbooks

No textbook is required. I recommend these books; my lectures draw on the first and fourth.

1. *SQL Queries for Mere Mortals,* ***Fourth Edition*** by John L. Viescas.
2. *Learning SQL,* ***Third Edition*** by Alan Beaulieu.
   It covers everything in this class, in a different order, and is **free online via the UCLA Library**.
3. *SQL for Smarties,* ***Fifth Edition*** by Joe Celko.
   This is an advanced book for SQL and it is highly recommended.
4. *Database System Concepts,* ***Seventh*** *Edition* by Silberschatz, Korth and Sudarshan.
   Theoretical and the most advanced of these, with much beyond SQL. It was formerly the course textbook.

Grading

Attendance & Participation	15%	Attendance via iClicker, course code UIDV
In-Class Quizzes	40%	Expect 5 or 6; lowest dropped
Final Exam	45%	**12/7 4:00–6:00pm in Anderson Hall G313**

Attendance and Participation

Attendance in lecture is required. Students are permitted one unjustified absence before a penalty adjustment is made to this portion of the grade. A quiz missed for any reason, **including a justified absence**, becomes your dropped quiz, leaving no drop for a later illness or emergency. Justified absences must be for medical, family, or unavoidable work-related reasons only and are at the sole discretion of the instructor. Attendance will be taken with iClicker, though we may switch to another tool at a later date and instructions will be provided if that happens. You will also be expected to complete certain tasks which will also count for credit in this category.

Problem Sets

There will be problem sets provided approximately each week. Some of the exercises involve writing SQL queries, some involve using AI effectively, and others are conceptual. This year, we will also work on some LeetCode problems as practice for job interviews. These assignments are **not graded**; treat them as preparation for the quizzes and final exam. You will work with real datasets from a variety of sources. In some cases, you may start the problem sets in lecture and finish at home.

Quizzes

There will be an in-class quiz approximately each week depending on our pace. These quizzes are closed-book, closed-note, closed-computer and will be taken with pencil and paper.

The lowest quiz score is dropped automatically to accommodate personal emergencies, illness, unavoidable travel, or an off day. As a result, no make-up quizzes will be administered for any reason, including justified absences.

Exams

There are two major assessments in this course. The diagnostic is graded for completion only; the final exam is graded for correctness.

Diagnostic Assessment: At the first lecture, everyone takes a diagnostic (multiple choice and free response on databases and SQL). You only need to **complete** it; your score is recorded but not reported or counted. Completing it counts toward your Attendance & Participation grade.

Final Exam: Cumulative, consisting of both diagnostic items and augmentation items that go beyond the diagnostic level, based on the course outcomes posted on BruinLearn. More information on format to follow. There are no make-up exams except where required by MSBA, Anderson or University policy. Typically the only recourse for missing a final exam due to an emergency is to request an Incomplete; however there are strict policies on when an Incomplete can and cannot be assigned. Otherwise, the final score is 0 and a final grade of "F" is submitted.

Request for Regrades

- Submit requests **within 1 week** of grade release.
- Use **Gradescope** (if applicable), otherwise a **private Piazza post**.
- Limit your request to **150 words**. Clearly state which part you believe is graded incorrectly and why.
- Regrade requests must identify a specific grading error; requests seeking reconsideration of the rubric or the amount of a correctly applied deduction will not be considered.
- Queries that produce the correct output but are **conceptually incorrect** will not receive full credit.
  - Examples: answers that work only by coincidence, omit required conditions, or violate problem constraints.
- Final regrade decisions rest with the instructor.

Final Grades

Final grade cutoffs are determined by the instructor based on demonstrated mastery of the course outcomes. The distribution of course scores may be considered when evaluating appropriate cutoffs, but grades are not assigned according to predetermined percentages or class rank. Your focus should be on learning the material as best you can. Every student who receives a passing grade in this class should be able to write SQL queries to solve the most common types of analytical problems. For more course outcomes, see BruinLearn.

Expectations

1. If you have questions about course content, ask them ASAP, either in class, discussion, office hours or Piazza.
2. If you encounter a technical issue, do not wait to get help. Please research first, then post on Piazza or come to office hours (if soon).
3. If you have a personal matter, please do not wait to contact Prof. Rosario in person, office hours or email.
4. Use the course staff, your peers, search and AI to help you (unless instructed not to by the instructor).
5. Assume best intentions – and I will do the same.
6. Behavior in lecture should be professional. Disruptive behavior that distracts other students or the instructor will not be tolerated, and I may ask you to leave the lecture. One example: Side conversations (including group chats) or repeated, disruptive laughter unrelated to course content, regardless of volume.

Tentative Schedule

Below is a tentative schedule of this course. **This schedule is subject to change.**

Week	Lecture	Date	Topic
1	1	September 28	Introduction and Diagnostic Exam
2	2	October 5	The Relational Model: Schemas and Keys (DDL/DML)
3	—	October 12	**No Class – Career Week**
4	3	October 19	Querying a Single Table and Aggregations
5	4	October 26	Introduction to Joins
6	5	November 2	Outer Joins and Subqueries
7	6	November 9	CTEs and Window Functions
8	7	November 16	Query Optimization and Indexing
9	—	November 23	**No Class – Thanksgiving Week**
10	8	November 30	Database Design & Architecture
Finals	—	December 7	**Final Exam, 4:00–6:00pm, Anderson Hall G313**

Academic Integrity

The UCLA Anderson School of Management and the instructor maintain strict standards of academic integrity under the UCLA Student Conduct Code. Because all evaluative coursework in this course consists of closed-book, in-person, handwritten assessments, the following operational boundaries are strictly enforced:

In-Class Assessments (Quizzes & Exams):

- All quizzes, the diagnostic assessment, and the final exam are strictly closed-book, closed-note, and individual.
- During an assessment, all electronic devices (including laptops, tablets, phones, and smartwatches) must be powered off and stowed in a bag away from your desk. Using or having out any device, notes, or other unauthorized aid will result in referral to the Dean of Students.

Generative AI & LLM Usage:

- AI tools (e.g., ChatGPT, Claude, Copilot) may be used *only* for study and ungraded problem sets.
- No AI, code completion, or search tools are permitted during in-class assessments.

Protection of Course Materials:

- Photographing, transcribing, distributing, or publicly posting any quiz, diagnostic, or exam prompt—including uploading to public repositories, course sharing platforms, or group communication channels—is prohibited and may be referred through the applicable UCLA academic-integrity/student-conduct process.

All students must complete the mandatory Academic Integrity Agreement on BruinLearn prior to receiving credit for any in-class assessment.